

Website Analytics Dashboard

Implementation playbook for the OMGEXP admin **/analytics** page — reusable on other sites.
Reference stack: Next.js + GA4 Data API + GTM on the public marketing site.
Version: July 2026

1. Architecture

The dashboard is **not** an embedded Google Analytics UI. It is a custom admin page that pulls aggregated metrics from the **GA4 Data API** using a Google Cloud service account, then renders charts with Recharts.

Visitor → Marketing site (GTM snippet) → dataLayer events → GTM → GA4 property
Staff → /analytics page → GET /api/analytics/ga (Bearer auth) → ga-data.ts → GA4 Data API → JSON → charts
Contact form → attribution in localStorage → Supabase (CRM) – separate from GA4, not shown on this dashboard yet

Two halves: (A) code in your repo embeds GTM and pushes events; (B) configuration in GTM UI + GA4 Admin (tags, triggers, custom dimensions, referral exclusions) is **not** in git.

2. Code-side setup (portable)

2.1 Environment variables (Tr backend)

Variable	Required	Description
GA4_PROPERTY_ID	Yes	Numeric property ID (GA4 Admin → Property settings)
GA_SERVICE_ACCOUNT_EMAIL	Yes	Service account email from Google Cloud
GA_SERVICE_ACCOUNT_PRIVATE_KEY	Yes	PEM key; use \n for newlines in env
MARKETING_GA_HOSTNAME	No	Default <code>www.omgcargo.tech</code> — filters event-scoped reports
GA_PUBLIC_PATH_PREFIX	No	Default <code>/site</code> — legacy marketing paths on the staff app domain
NEXT_PUBLIC_GTM_ID	No	GTM on public pages (separate from dashboard API)

2.2 Key source files

- `src/lib/ga-range.ts` — date windows using GA4 relative dates (`7daysAgo ... yesterday`)
- `src/lib/ga-data.ts` — Data API queries, site filter, lead events
- `src/app/api/analytics/ga/route.ts` — admin-only API + 30-minute in-memory cache
- `src/app/(main)/analytics/page.tsx` — dashboard UI

2.3 Site filter logic

Reports include traffic where **either**:

- `hostName` = `MARKETING_GA_HOSTNAME` (new Astro site at `www.omgcargo.tech`), or
- `pagePath` begins with `GA_PUBLIC_PATH_PREFIX` (legacy `/site/*` on `cargo.omgexp.com`)

Staff-only routes (e.g. `/dashboard`, `/login`) are excluded unless they match the above.

2.4 Marketing site events (dataLayer)

Push these from the public site after GTM loads:

Event	Parameters	When
<code>generate_lead</code>	<code>form_name</code> , optional <code>transaction_id</code>	After contact API returns success (not on click)
<code>cta_click</code>	<code>cta_label</code> , <code>cta_location</code>	Primary CTA clicks
<code>contact_click</code>	<code>contact_method</code> , <code>contact_value</code>	tel: / mailto: links

3. Manual setup (GTM + GA4 Admin)

1. Create a GA4 property and web data stream for your marketing domain.
2. Create a GTM container; add GA4 Configuration tag with your Measurement ID (G-xxxx).
3. Add Custom Event triggers for `generate_lead`, `cta_click`, `contact_click`.
4. Add GA4 Event tags mapping `dataLayer` parameters to event parameters.
5. In GA4 Admin → Custom definitions, register event-scoped dimension **`form_name`** (parameter `form_name`). Without this, "By form" breakdown fails.
6. Enable **Google Analytics Data API** in Google Cloud; create a service account; download JSON key or copy email + private key to env.
7. GA4 Admin → Property access management → add service account as **Viewer**.
8. GA4 Admin → Data streams → Configure tag settings → define **unwanted referrals** (e.g. your own domain) to fix self-referral like `omgcargo.tech / referral1`.
9. Mark `generate_lead` as a key event in GA4 if you want it in standard reports.

4. What each dashboard metric means

UI label	GA4 source	Means	Does NOT mean
Users / Sessions / Page views	Filtered traffic totals	Marketing + legacy /site paths	All staff app usage
Lead events	eventCount for generate_lead	Successful form submits tracked in GA4	Unique people; CRM row count
Top entry pages	pagePath + entrances	Pages where sessions started	Same as landingPage dimension (session-scoped)
First-touch source	firstUserSourceMedium	First acquisition source on the GA4 property	First visit to marketing site only
Top sources	sessionSourceMedium	Source/medium for each session	Last-click attribution only

Period-over-period % compares equal-length windows ending **yesterday** (today excluded — GA4 processing lag).

5. Accuracy limits

- **Processing lag:** Standard reports lag 24–48 hours; Realtime/DebugView are immediate.
- **Undercount:** Ad blockers, no-JS, Safari ITP — typically 10–30% of sessions missing client-side.
- **Lead events vs CRM:** GA4 counts events; Supabase stores submissions with attribution JSON — two systems, two numbers.
- **Double submits:** eventCount increments per event; GA4 dedupes transaction_id only for purchase, not generate_lead.
- **Legacy /site traffic:** If old marketing routes stay live, metrics mix two sites until you redirect or narrow filters.
- **Self-referral:** Cross-domain hops without referral exclusion inflate “referral” and split sessions.

End-to-end verification not yet done: Submit one real test lead via the live contact form, then confirm (1) GA4 DebugView shows generate_lead, (2) dashboard Lead events increments after lag, (3) Supabase contact_submissions.attribution has UTM/click IDs.

6. Verification checklist

1. GTM Preview → submit test form → see generate_lead with form_name.
2. GA4 DebugView → same event within seconds.
3. GA4 Realtime → active user on marketing hostname.
4. Admin /analytics → KPIs load (not “Analytics not configured”).

5. If a section shows **Data unavailable** with an error string, fix GA4 config (usually missing custom dimension).
6. Run `node --experimental-strip-types scripts/ga-range.test.mjs` after changing date logic.
7. Compare lead events (GA4) vs contact rows (Supabase) over the same week — expect difference, not parity.

7. Appendix

7.1 Env template

```
GA4_PROPERTY_ID=123456789
GA_SERVICE_ACCOUNT_EMAIL=analytics-read@project.iam.gserviceaccount.com
GA_SERVICE_ACCOUNT_PRIVATE_KEY="-----BEGIN PRIVATE KEY-----\n...\n-----END PRIVATE KEY-----\n"
MARKETING_GA_HOSTNAME=www.example.com
GA_PUBLIC_PATH_PREFIX=/site
NEXT_PUBLIC_GTM_ID=GTM-XXXXXXX
```

7.2 PEM key troubleshooting

If the API returns `OpenSSL DECODER routines::unsupported`, the private key env is malformed. The codebase normalizes quotes, literal `\n`, and Windows `\r` — but the key must still be a valid PEM block.

7.3 Legacy /site vs new domain

OMG currently serves marketing on both `www.omgcargo.tech` (Astro) and `/site/*` on the Next app. Top pages may show `/site/login` alongside `/`. Options: 301 legacy to new domain (SEO + cleaner analytics), or set `GA_PUBLIC_PATH_PREFIX` to a path that no longer receives traffic once migrated.

7.4 PDPA note

GTM/GA4 cookies require disclosure in your cookie policy. Attribution keys in `localStorage` should be listed there too. Consent Mode v2 is recommended for new deployments in regulated markets.